

Project Quantum: Technical Architecture

1. Overview

Project Quantum is our next-generation caching layer. It replaces the legacy Memcached cluster with a globally distributed Redis enterprise deployment.

The migration is scheduled for Q3 2024.

2. Performance Targets

The primary goal is to reduce P99 latency from 45ms to under 10ms for read operations.

Write operations are expected to maintain a P95 latency of 15ms.

3. Incident Response

If the caching layer fails, the application will fallback to direct database queries. The database cluster is managed by the DBA team led by Sarah Connor.

If database load exceeds 80%, circuit breakers will automatically shed non-critical traffic.